ChatGPT

Adapting Audio Foundation Models for Heart Sound Analysis (Shenoi et al., 2025)

Shenoi *et al.* used OPERA-CT as a base encoder for cardiac auscultation tasks [1] [2]. Domain: **Physiological (heart sounds)**. Datasets: four public heart-sound datasets including CirCor '22 (murmur and outcome labels), PhysioNet/CinC 2016 (normal/abnormal), PASCAL, and ZCHSound. Method: OPERA-CT (pretrained on respiratory audio) was used with linear probing (LP) and full fine-tuning (FT), and further adapted by *continued pretraining* (CP) on heart-sound data [3] [4]. Outcomes: Baseline OPERA-CT (LP) achieved macro-AUROC ~0.831 on the PhysioNet task, which improved to 0.927 after full FT [2]. Fine-tuning increased AUROC by up to ~12% over LP [5], though at the cost of reduced cross-task generalization. Interestingly, in-corpus CP boosted LP performance, but gains often vanished after FT [6]. Overall, OPERA-CT provided strong representations for murmur/outcome classification (e.g. baseline AUROC 0.831 → 0.927 after FT on one task) [2]. *Link:* Shenoi et al. (CINC 2025).

Harnessing Pretrained Audio Models for Mood Awareness in Breathing (Shenoi *et al.*, 2025)

Shenoi *et al.* also evaluated OPERA-CT as a feature extractor for mood prediction from breathing sounds [7]. Domain: **Affective computing (breathing/voice audio)**. Dataset: an in-lab dataset (N=29) of 3-minute breathing exercises, with subjective agitation scores before/after. Method: Pretrained OPERA-CT (and VGGish) were used to extract audio embeddings (4-second windows), followed by classifiers (DNN, RF, XGBoost) to predict "low" vs "moderate/high" agitation. Outcomes: Using OPERA-CT features gave good performance (weighted F1 $\gtrsim$0.75), but VGGish (trained on general AudioSet) performed slightly better. For example, VGGish+XGBoost achieved F1≈0.83, outperforming OPERA-CT [7]. The authors note that automated features (VGGish/OPERA) substantially outperformed manual (OpenSMILE) features in predicting agitation [7]. *Link:* Shenoi *et al.* (CHI'25 workshop).

Cough-based Respiratory Disease Diagnosis (Wang *et al.*, 2024)

Wang *et al.* fine-tuned OPERA-CT for cough sound disease screening [8]. Domain: **Respiratory health (COVID-19, COPD)**. Datasets: a new large "LucaCough" dataset plus two benchmarks (UK COVID-19 cough, COUGHVID). Method: Various pretrained models were compared. OPERA-CT (and variants) pretrained on respiratory sounds was *fine-tuned* on the cough tasks. Outcomes: Fine-tuned OPERA-CT achieved AUROC ≈85.2% on LucaCough, 69.4% on UK-COVID, and 58.4% on COUGHVID [8]. These results trailed those of models pretrained on large general-audio corpora (e.g. PaSST, EAT). The authors found that OPERA-CT (self-supervised on ~400 h of resp audio) underperformed larger models, likely due to its smaller capacity and narrower pretraining data [9]. *Link:* Wang *et al.* (arXiv 2024).

Industrial Machine Sound Classification (Zhao *et al.*, 2025)

Zhao *et al.* used OPERA-CT as a frozen feature encoder for machine fault classification [10]. Domain: **Industrial machine audio**. Dataset: DINOS – a new benchmark (~1,093 h, various CNC and additive machines). Method: OPERA-CT and OPERA-GT (trained on stethoscope-recorded respiratory sounds) were applied via linear-probe to classify machine operating conditions. Outcomes: OPERA-CT performed extremely well on stethoscope-recorded tasks – often near-perfect. For example, on one machine (Renishaw-L), OPERA-CT achieved F1=1.000 (100%) [10]. Across machines, OPERA-CT F1 scores were $\gtrsim$0.94 (Table 3) [10], far above classical features (OpenSMILE ~0.02–0.63). OPERA models outperformed general audio embeddings in these stethoscope scenarios [11]. However, performance dropped on ambient mic recordings, suggesting domain mismatch [12]. The newer IMPACT model further surpassed OPERA (F1>0.96 on all tasks) by pretraining on DINOS, but OPERA-CT still provided strong baselines [11] [10]. *Link:* [Zhao *et al.* (arXiv 2025)].

RespLLM: Multimodal Respiratory Health LLM (Zhang *et al.*, 2024)

Zhang *et al.* introduced **RespLLM**, which uses OPERA-CT as the audio encoder in a multimodal LLM framework [13]. Domain: **Respiratory health (multimodal)**. Data: Five heterogeneous datasets combining patient demographics, symptoms (text) and respiratory audio. Method: OPERA-CT processes 8 s of audio into 64 patch embeddings [13]; these are fused (via cross-attention) with a biomedical LLM (OpenBioLLM-8B) in an instruction-tuned pipeline. This joint model is trained on curated tasks for diagnosis. Outcomes: RespLLM significantly beat strong baselines. On held-out (in-domain) tasks, RespLLM's average AUROC was 0.8072 vs 0.7717 for the best prior method (a +4.6% gain) [14]. In zero-shot transfer to new datasets/tasks, it achieved average AUROC 0.6547 vs 0.6070 (+7.9%) [15]. (For instance, it could predict "asthma" on a new dataset with AUROC ~0.5865, where baselines failed.) This shows that OPERA-CT features, combined with LLM fusion, yielded more generalizable respiratory diagnostics [14] [15]. *Link:* [Zhang *et al.* (arXiv 2024)].

Sources: Each paper above is cited with key results from the original text [1] [2] [5] [7] [13] [14] [15] [8] [10] [11]. Links (in parentheses) point to the cited papers.

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